Data Structures

- Data structure is a particular format of organizing data such that it can be used efficiently.
- A data structure can be a built-in data type or a user-defined abstract data type.

- Examples:
  - **int**: addition, subtraction, multiplication, division
  - **Stack**: push, pop, top, isempty, isfull
  - **Queue**: enqueue, dequeue, isempty, isfull
  - **Tree**: insert, delete, search
  - **Heap**: insert, delete, min

Design of a Data Structure

- Specification
  - A set of data
  - Domain of data
  - Operations to be performed on data
- Design
  - Algorithms/functions for the operations
- Goals of Design: efficiency

Implementation of a Data Structure

- Representation of the data using built-in data types of the programming language such as int, double, char, strings, arrays, structs.

- Representation of the data using user-defined data types (abstract data types-ADT): stack, queue, trees, graphs.
  - Data
  - Domain
  - Operations
  - Behavior of operations

Performance Analysis

- Determining an estimate of the time and memory requirement of the algorithm.
  - Time estimation is called time complexity analysis
  - Memory size estimation is called space complexity analysis.

- We primarily focus on time complexity.

Example of Time Complexity Analysis

Finding a maximum of an array of *n* integers.

```
int m = a[0];
for i = 1 to n-1 do
   if (a[i] > m)
         m = a[i];
return m;
```

No. of operations ?